A Comparative Analysis of Classical Machine Learning Methods and Variational Quantum Methods for Breast Tumor Classification

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February 26, 2026

Outline

Introduction

Our goals:

- Tumor Classification using Classical ML methods
- Tumor Classification using Variational Quantum methods
- Comparison of the results

Thesis outline

Workflow of the master thesis:

- Classical Machine Learning Methods Introduction
- Quantum Computing and Quantum Machine Learning Basics
- EDA & Classification using Classical ML methods and rnCV
- Classification using PCA and VQCs
- Comparison of the Classical and Quantum approaches

Classical Machine Learning Basics

Two types of ML:

- Supervised (Labeled Datasets)
- Unsupervised (Hidden patterns, no labels)

Unsupervised ML:

- Clustering
- Anomaly Detection
- Dimensionality Reduction

Supervised ML:

- Regression
- Classification

We will focus on **Classification**

Binary Classification with Classical ML Methods

Binary Classification: Choose between **two categories**

Classical ML methods:

- **Logistic Regression**
- **Gaussian Naive Bayes** (GNB)
- **Linear Discriminant Analysis** (LDA)
- **Support Vector Machines** (SVM)
- **Random Forests**
- **Light Gradient Boosting Machine** (LightGBM)

Quantum Computing and Quantum Machine Learning Basics

Key points:

- The Qubit
- Qubit Manipulation and Quantum Entanglement
- Quantum Gates
- Quantum Circuits
- Data Encoding
- Variational Quantum Classifiers

Quantum Mechanics and Quantum Computers

Quantum Theory or **Quantum Mechanics**:

- Mathematical **Framework** for the results of measurement of **quantum systems**

Quantum Systems:

- **Microscopic** entities where **Newtonian** mechanics **fail** (e.g hydrogen atom)

Quantum Computer:

- Computational device that **exploits quantum mechanical effects** (superposition,entanglement) to carry out **operations on information**
- Uses **Qubits** instead of bits
- **Quantum characteristics** of particles can be **utilized**
- **Quantum processes** can **manipulate** this information

The Qubit

- Fundamental building blocks of quantum computers
- The quantum mechanical analogue of a classical bit
- Created by manipulating and measuring systems that exhibit quantum mechanical behavior
  - **Superconducting qubits**: **superconducting** materials at **cryogenic** temperatures, offer **rapid** gate speeds and **precise** control
  - **Trapped ion qubits**: individual **ions confined** by electromagnetic fields, **long coherence** times and **high-fidelity measurements**, slower than **superconducting**
  - **Quantum dots**: tiny **semiconductors** capture a **single electron**, potential for **scalability**
  - **Photons**: particles of **light**, encode quantum information and transmit via **optical fibers**, applications in quantum **communication** and **cryptography**

The Qubit, more formally

A two-state quantum system defined by the **basis vectors** $|0\rangle$ and $|1\rangle$

A qubit may exist in a **superposition**, a mathematical **linear combination** of both basis states, $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$

$|0\rangle = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$, $|1\rangle = \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$, α, β probability amplitudes with complex values

Hermitian conjugate: $\langle\psi| = \begin{pmatrix}\alpha^* & \beta^*\end{pmatrix}$

$|\cdot\rangle$ is called "**ket**", $\langle\cdot|$ is called "**bra**"

Upon **measurement**, Qubit **collapses** with $P(|0\rangle) = |\alpha|^2$ and $P(|1\rangle) = |\beta|^2$

Thus, it must hold $|\alpha|^2 + |\beta|^2 = 1$ or $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$

If not, then we do $|\tilde{\psi}\rangle = \frac{|\psi\rangle}{\|\psi\|}$

A qubit is defined as a **vector** residing in a **two-dimensional complex Hilbert** space $\mathbb{C}^2$

Bloch sphere

$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$
$|\psi\rangle = |0\rangle$ occurs at $\theta = 0$
$|\psi\rangle = |1\rangle$ occurs at $\theta = \pi$

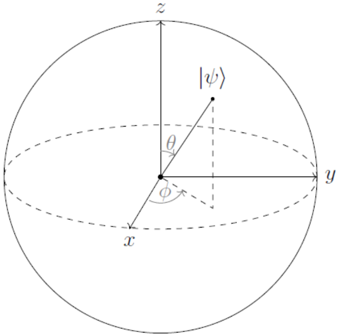


Figure: A qubit can be mapped to a point on the surface of this unit-radius sphere ($r = 1$), uniquely determined by the angular coordinates θ and ϕ

Operators

We transition qubits through **different states** with the use of **operators**

- $\hat{A}\,|\psi\rangle = |\phi\rangle$, $\hat{A}\,\langle\psi| = \langle\phi|$
- Vital set of operators called **Pauli** operators:
  - $\hat{I}|\psi\rangle = |\psi\rangle$
  - $\hat{X}|0\rangle = |1\rangle$, $\hat{X}|1\rangle = |0\rangle$ (*NOT* operator)
  - $\hat{Y}|0\rangle = -i\,|1\rangle$, $\hat{Y}|1\rangle = i\,|0\rangle$
  - $\hat{Z}|0\rangle = |0\rangle$, $\hat{Z}|1\rangle = -\,|1\rangle$

We can also express an operator as an outer product

- $(|\psi\rangle\,\langle\phi|)\,|\chi\rangle = |\psi\rangle\,\langle\phi|\chi\rangle = \langle\phi|\chi\rangle\,|\psi\rangle$

Operators as Matrices

Another method for representing an operator is through matrix notation

For an n-**dimensional** vector space $\rightarrow n \times n$ **matrices**. We focus on $\mathbb{C}^2$, so 2×2 **matrices**

$$\hat{A} = \begin{pmatrix} \langle 0|\hat{A}|0\rangle & \langle 0|\hat{A}|1\rangle \\ \langle 1|\hat{A}|0\rangle & \langle 1|\hat{A}|1\rangle \end{pmatrix} \tag{1}$$

Pauli Operators as Matrices

Following Equation 1 we get that:

$$\hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{2}$$

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{3}$$

$$\hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{4}$$

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{5}$$

Hermitian & Unitary Operators

A given operator $\hat{A}$ is classified as **Hermitian** iff:

$$\hat{A} = \hat{A}^{\dagger} \tag{6}$$

- $\hat{A}^{\dagger}$ denotes the Hermitian adjoint $\rightarrow$transposing the matrix and taking the complex conjugate of every entry
- Hermitians possess **real eigenvalues** and their **main diagonal** consists solely of **real** numbers.
- Therefore they represent **physical observables**

A given operator $\hat{U}$ is classified as **Unitary** iff:

$$\hat{U}\hat{U}^{\dagger} = \hat{U}^{\dagger}\hat{U} = \hat{I} \tag{7}$$

- They characterize the **temporal evolution** of a quantum state

The **Pauli** operators are **both** the Hermitian and Unitary

Outer product

- To explore more sophisticated and advantageous quantum algorithms, we need systems of **multiple interacting qubits**
- Construction of a **composite** Hilbert space
- The mathematical operation used is the Kronecker or tensor product ($\otimes$)

If H_1 and H_2 two Hilbert spaces with dimensions N_1 and N_2, then we can form a composite Hilbert space H with dimension $N_1 N_2$, defined as:

$$H = H_1 \otimes H_2 \tag{8}$$

States derived from tensor product

The tensor product can be employed to construct a state residing within the composite space H:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle \tag{9}$$

For two-dimensional subspaces, the following procedure is employed to calculate the product:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix} \tag{10}$$

Usually, the $\otimes$ symbol is omitted; for example, $|\phi\rangle \otimes |\chi\rangle$ may be written as $|\phi\rangle |\chi\rangle$, or even more compactly as $|\phi\chi\rangle$.

Basis vectors and operators of a composite system

If $|u_i\rangle$ and $|v_i\rangle$ are the basis vectors of H_1 and H_2, the resulting basis for the joint space H is:

$$|w_i\rangle = |u_i\rangle \otimes |v_i\rangle \tag{11}$$

We can define operators that act upon our composite system. If $|\phi\rangle \in H_1$ and $|\chi\rangle \in H_2$, and A acting on $|\phi\rangle$ and B acting on $|\chi\rangle$. A new operator, which acts on a state $|\psi\rangle \in H$ is:

$$(A \otimes B)\,|\psi\rangle = (A \otimes B)(|\phi\rangle \otimes |\chi\rangle) = (A\,|\phi\rangle) \otimes (B\,|\chi\rangle) \tag{12}$$

The matrix representation is computed as:

$$A \otimes B = \begin{pmatrix} \alpha_{11}B & \alpha_{12}B \\ \alpha_{21}B & \alpha_{22}B \end{pmatrix} = \begin{pmatrix} \alpha_{11}\beta_{11} & \alpha_{11}\beta_{12} & \alpha_{12}\beta_{11} & \alpha_{12}\beta_{12} \\ \alpha_{11}\beta_{21} & \alpha_{11}\beta_{22} & \alpha_{12}\beta_{21} & \alpha_{12}\beta_{22} \\ \alpha_{21}\beta_{11} & \alpha_{21}\beta_{12} & \alpha_{22}\beta_{11} & \alpha_{22}\beta_{12} \\ \alpha_{21}\beta_{21} & \alpha_{21}\beta_{22} & \alpha_{22}\beta_{21} & \alpha_{22}\beta_{22} \end{pmatrix} \tag{13}$$

Quantum Entanglement

A composite system **cannot always** be expressed as a tensor product of individual states, only if the qubits are prepared separately and kept in isolation. If the qubits are permitted to **interact**, they form a single integrated closed system and are described as **entangled**:

- **Quantum Entanglement** is a fundamental phenomenon and a vital resource for quantum computation
- **Particles** sharing a common origin remain fundamentally **linked** regardless of the distance between them
- Any measurement or interaction of one particle instantaneously **influences** the state of its entangled **partners**
- Information can be transmitted **faster** than the speed of **light**

Quantum Gates

Transformations applied to qubit states are referred to as quantum gates, and a sequence of such operations forms a **quantum circuit**:

- **Quantum gates** represent unitary **operations** performed on **qubits**
- The terms "**gate**" and "**unitary operator**" are used **interchangeably**
- A gate processing n **inputs** and outputs is represented by a $2^n \times 2^n$ **matrix**
- A **single-qubit** gate is depicted as a 2×2 **matrix**
- A quantum gate acting on **two qubits** requires a 4×4 **matrix** representation

The most fundamental single-qubit operation is the quantum NOT gate, which is equivalent to the Pauli X operator. This gate effectively swaps the basis states. Represented as a matrix in the computational basis, we have:

$$\hat{U}_{NOT} = \hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{14}$$

Hadamard and Rotation gates

The Hadamard gate, which transforms the standard basis states into a balanced superposition

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{15}$$

Rotations of a qubit along the x, y, and z axes of the Bloch sphere:

$$\hat{R}_x(\theta) = e^{-i\theta \frac{X}{2}} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \tag{16}$$

$$\hat{R}_y(\theta) = e^{-i\theta \frac{Y}{2}} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \tag{17}$$

$$\hat{R}_z(\theta) = e^{-i\theta \frac{Z}{2}} = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{-i\frac{\theta}{2}} \end{pmatrix} \tag{18}$$

Any arbitrary single-qubit gate can be represented as a specific sequence of rotations around the z and y axes (Z-Y decomposition)

The *controlled-NOT* (*CNOT*) gate

Two-qubit gate where the target qubit remains unchanged if the control qubit is in the state $|0\rangle$; however, if the control qubit is in the state $|1\rangle$, the target qubit undergoes a bit-flip, as if a *NOT* gate had been applied

$$CNOT\,|00\rangle = |00\rangle$$

$$CNOT\,|01\rangle = |01\rangle$$

$$CNOT\,|10\rangle = |11\rangle$$

$$CNOT\,|11\rangle = |10\rangle$$

Its matrix representation is :

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{19}$$

Quantum Circuits

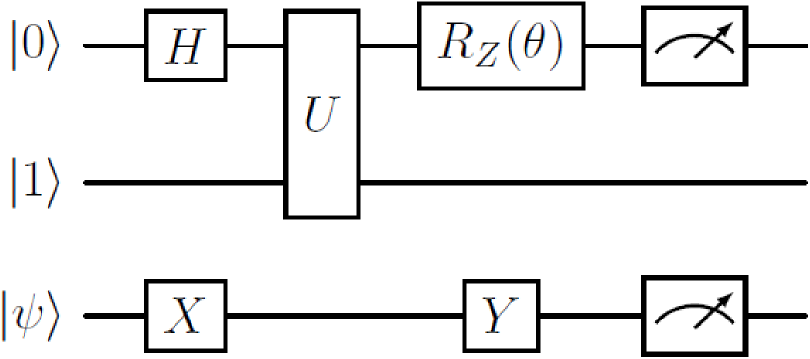


Figure: An example of a quantum circuit diagram

Input Encoding Techniques

Various techniques to **embed classical data** into an n-qubit quantum system:

- **Basis** encoding
- **Time-evolution** encoding
- **Hamiltonian** encoding
- **Amplitude** encoding

We will be using **Amplitude** encoding with **Amplitude-Efficient State Preparation** to encode 2^n features using n qubits

Variational Quantum Classifiers

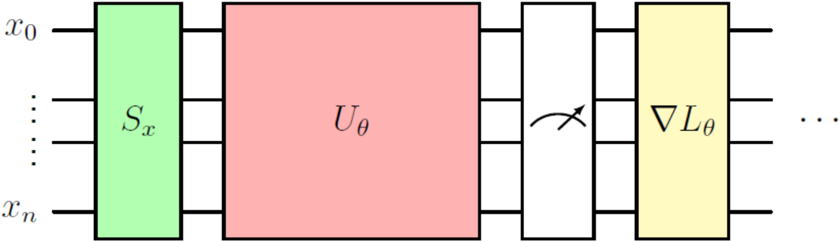


Figure: General structure of a variational quantum classifier

Breast Cancer

- **Most common** malignancy among women worldwide
- **36%** of all oncological cases
- An estimated **2.089 million** women diagnosed in 2018

Lifestyle related factors:

- **Unhealthy dietary** habits
- **Nicotine** use
- Chronic **stress**
- **Low** levels of **physical** activity

Other risk factors:

- Advanced **age**
- Family **history** of breast cancer
- Infection with **oncogenic** viruses
- **Early** onset of **menstruation** or **late menopause**
- **First** pregnancy at an **advanced** age or the **absence** of pregnancy
- Use of **hormone replacement therapy**

Fine Needle Aspiration

We will be working with samples derived from Fine Needle Aspirates

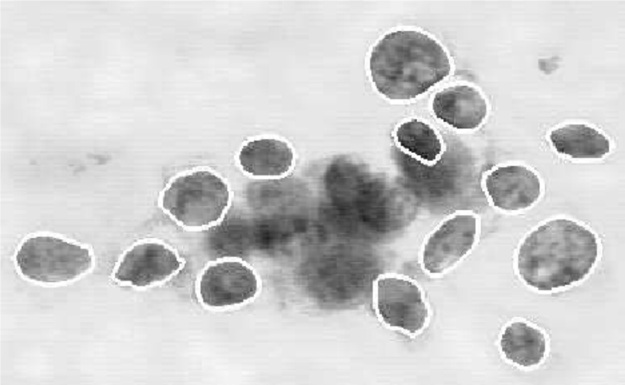


Figure: A magnified image of a malignant breast FNA with the visible cell nuclei having been outlined with the help of a curve fitting program.

Evaluation Metrics

Six models from before with **nine** key **metrics**:

- (Balanced) **Accuracy** ($\frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$)
- **Precision** ($\frac{TP}{TP+FP}$)
- **Recall** ($\frac{TP}{TP+FN}$)
- **F1-Score** ($2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP+FP+FN}$)
- **MCC** ($\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$)
- **ROC AUC** ($\int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$)
- **PR AUC** ($\int_0^1 \text{Precision}(\text{Recall})\, d(\text{Recall})$)
- **Specificity** ($\frac{TN}{TN+FP}$)
- **NPV** ($\frac{TN}{TN+FN}$)

Evalution Outline

Exploratory Data **Analysis** (feature distribution, class imbalance)

Finding the **best** model (80-20 spilt, rnCV, data cleaning)

Fine tune the winner (bootstrapping, 95% confidence intervals)

Exploratory Data Analysis

30 features with **569** entries

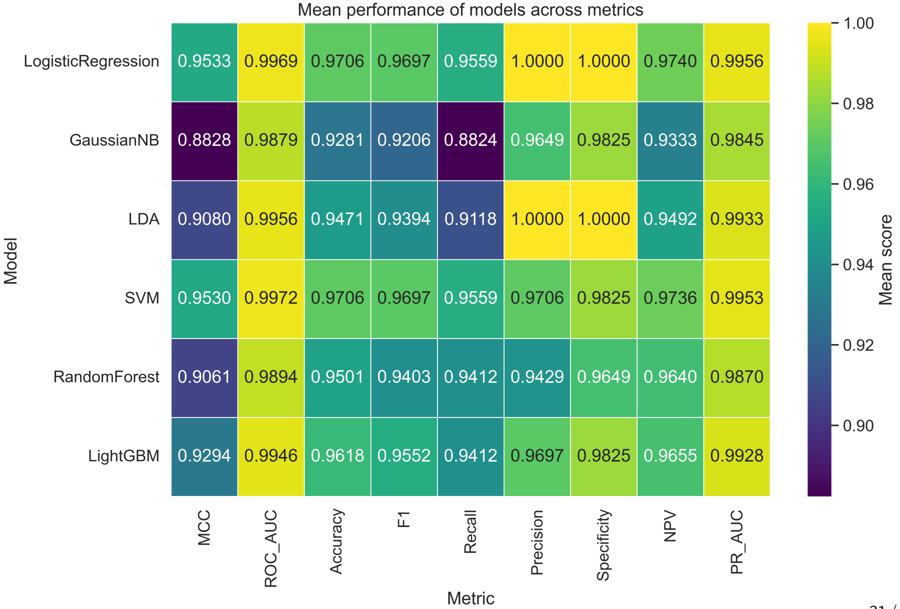
Class **imbalance**: **357** instances of **benign**, **212** instances of **malignant** (62.74% - 37.25%)

Majority of features exist on **different** scales, will **scale inside** rnCV

15 out of 30 features heavily **correlated** with **prognosis**

No correlation among features that we did not expect

Logistic Regression as the best model



Finetuning and bootstrapping

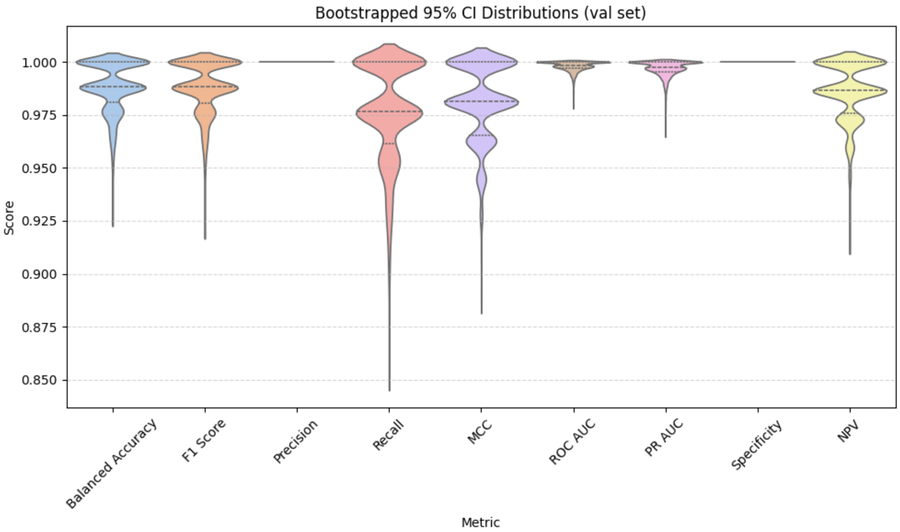


Figure: Violin plot of the confidence intervals

Quantum Experimental Framework

- **1**, **2**, **3**, **4** and **5** qubits with **2**, **4**, **8**, **16** and **30** features through **PCA**
- **60-20-20** train-validation-test split
- -**1** for **benign** and **1** for **malignant**
- Mean Squared Error (**MSE**) for **loss** function, **Nesterov** Momentum for **optimizer**
- Focus on **4** metrics: Accuracy, Recall, Precision, F1-score
- Circuit consisting of **3** parts: Data **encoding**, **Variational** and **Measurement**
- **Same** architecture for Variational: **Layers** consisting solely of **Rotations** and **CNOT**s
- **Two** variants of VQCs: **Unbiased** vs **Biased**
- **Set** number of training **epochs** for each **qubit**, we keep track of **time**
- **Average** over **10** runs, keep **track** of the **best** performers per variation

One qubit results

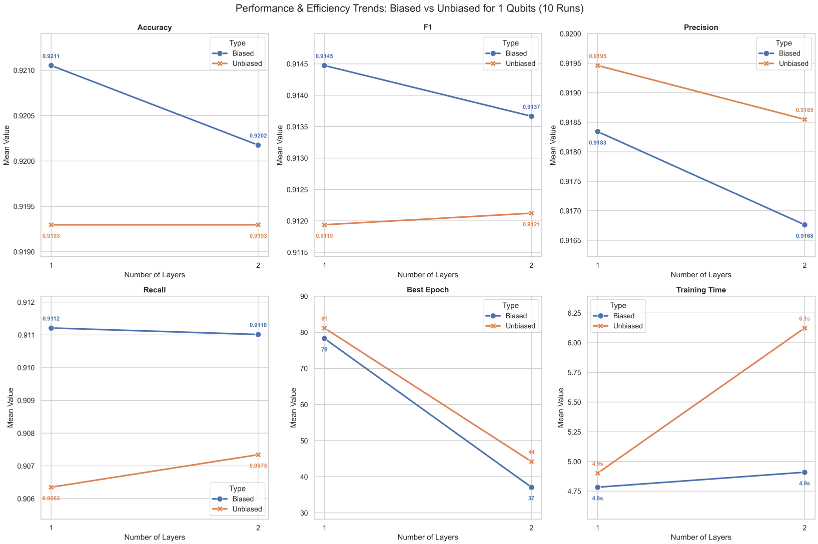


Figure: Performance and Efficiency Trends of Biased and Unbiased VQC for 1-Qubit

One qubit results, circuit of best unbiased VQC

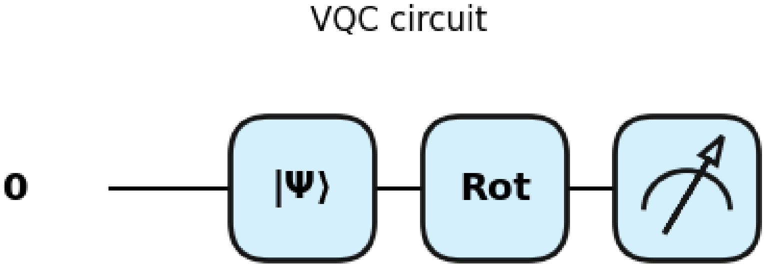


Figure: Circuit of the best unbiased 1-qubit VQC

One qubit results, loss evolution of best unbiased VQC

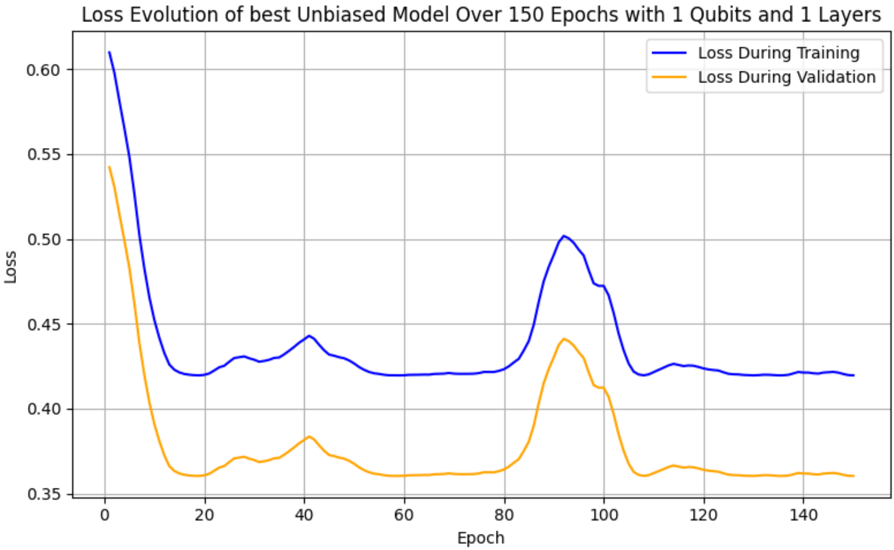


Figure: Evolution of the loss function of the best unbiased 1-qubit VQC

One qubit results, performance evolution of best unbiased VQC

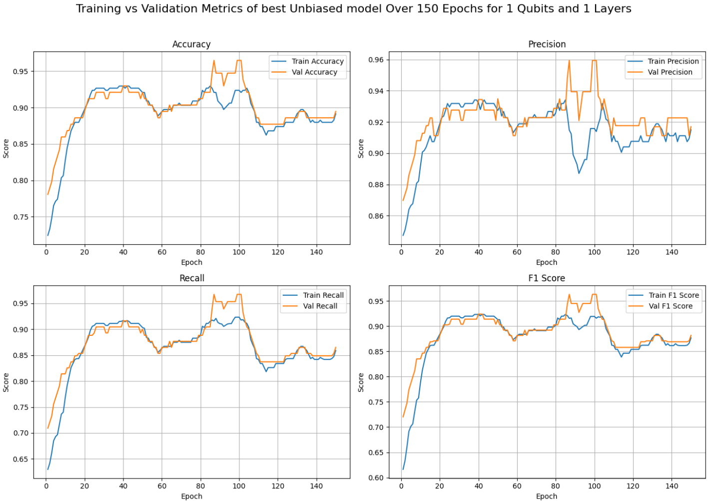
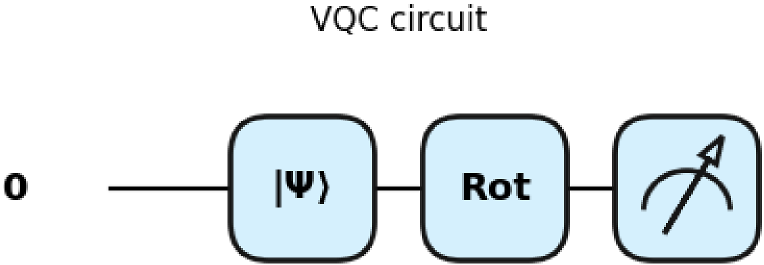


Figure: Validation and Training curves for the 4 key metrics of the best unbiased 1-qubit VQC

One qubit results, circuit of best biased VQC



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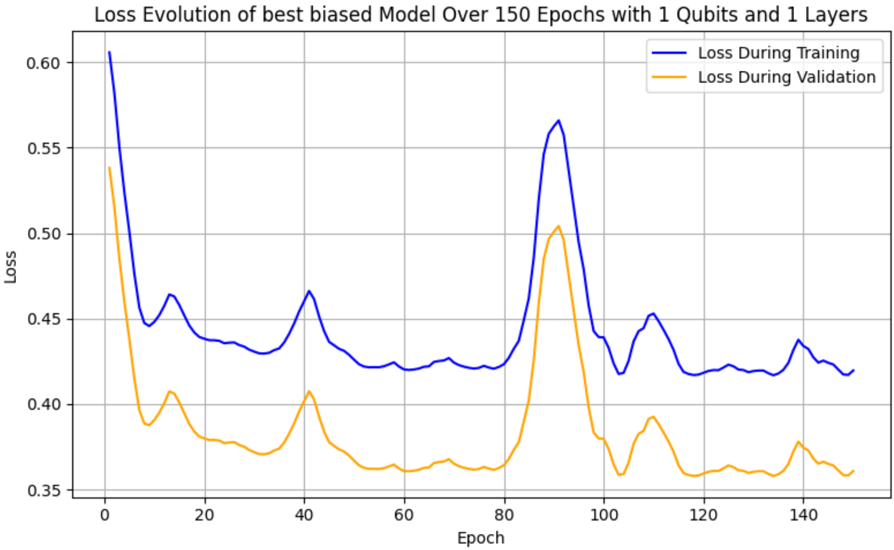

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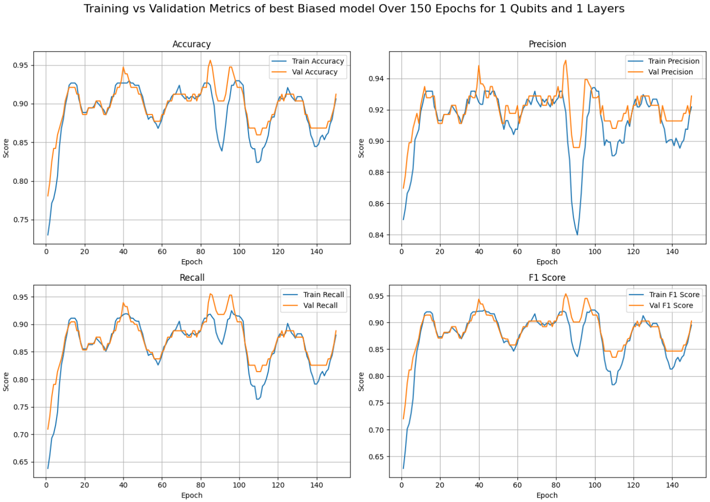

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Two qubit results

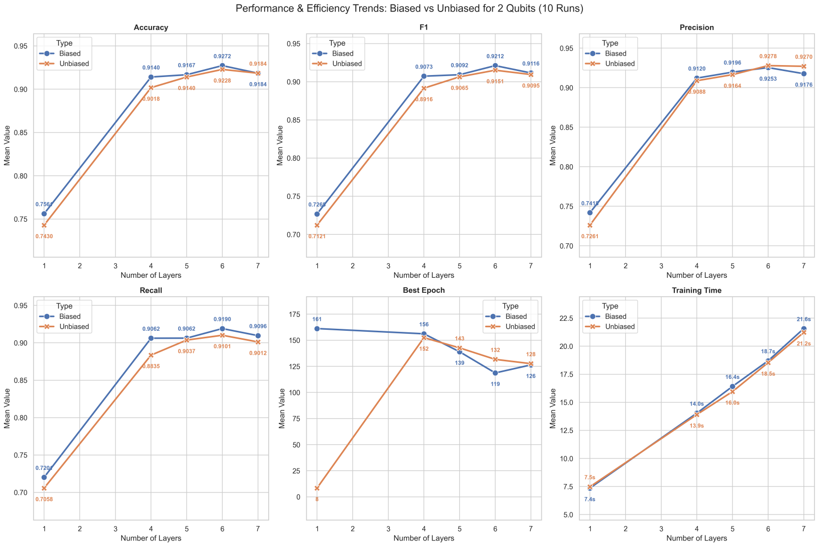


Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 2 Qubits

Two qubit results, circuit of best unbiased VQC

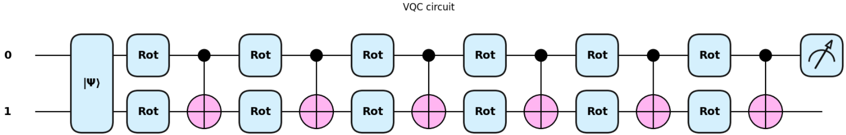


Figure: Circuit of the best 2-qubit unbiased classifier

Two qubit results, loss evolution of best unbiased VQC

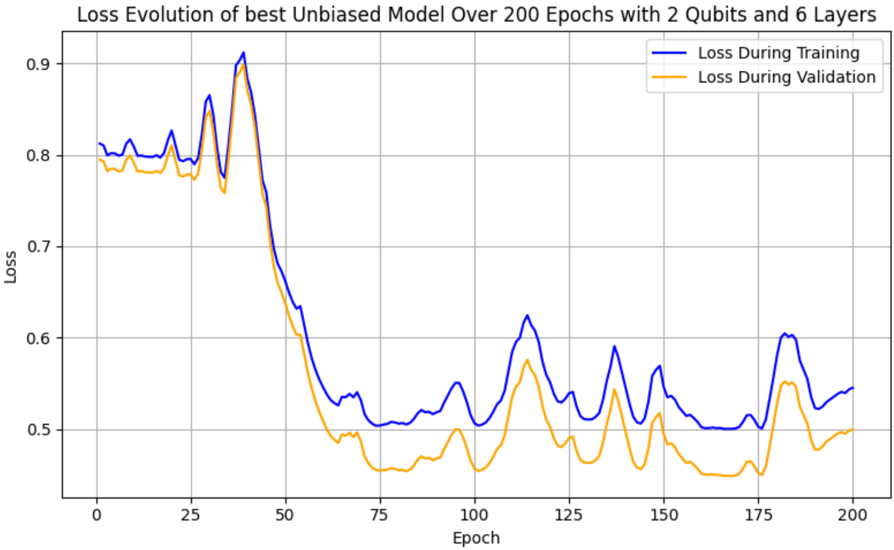


Figure: Evolution of the loss function of the best 2-qubit unbiased classifier

Two qubit results, performance evolution of best unbiased VQC

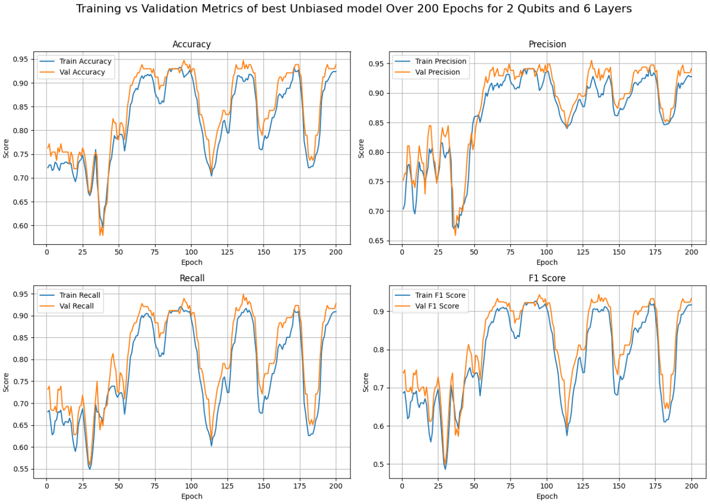


Figure: Validation and Training curves for the 4 key metrics of the best 2-qubit unbiased classifier

Two qubit results, circuit of best biased VQC

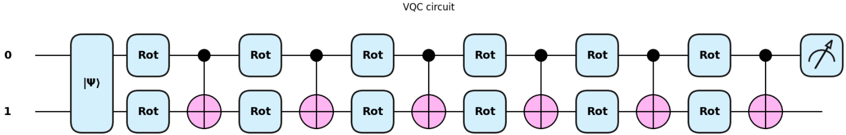


Figure: Circuit of the best 2-qubit biased classifier

Two qubit results, loss evolution of best biased VQC

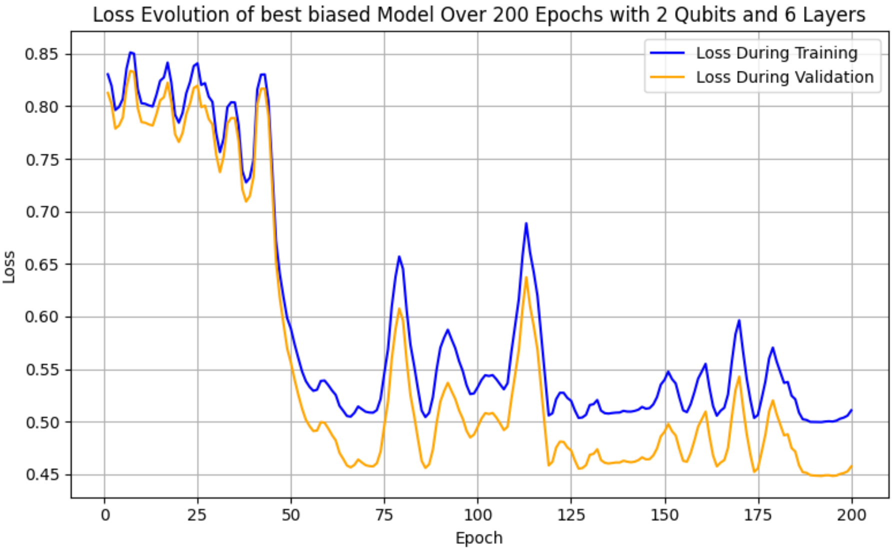


Figure: Evolution of the loss function of the best 2-qubit biased classifier

Two qubit results, performance evolution of best biased VQC

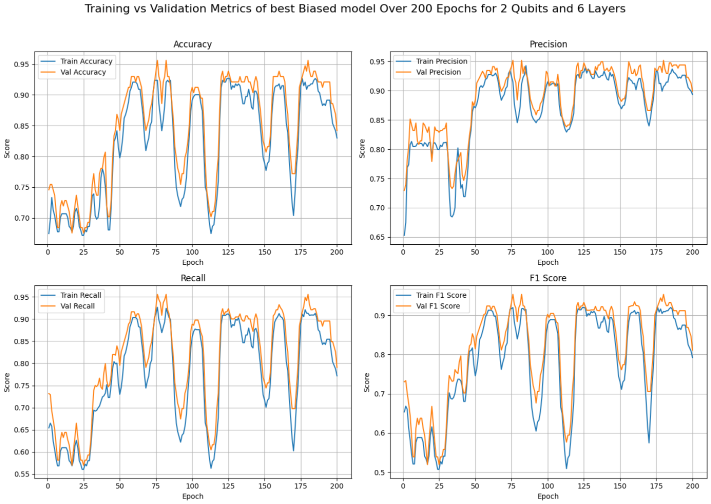


Figure: Validation and Training curves for the 4 key metrics of the best 2-qubit biased classifier

Three qubit results

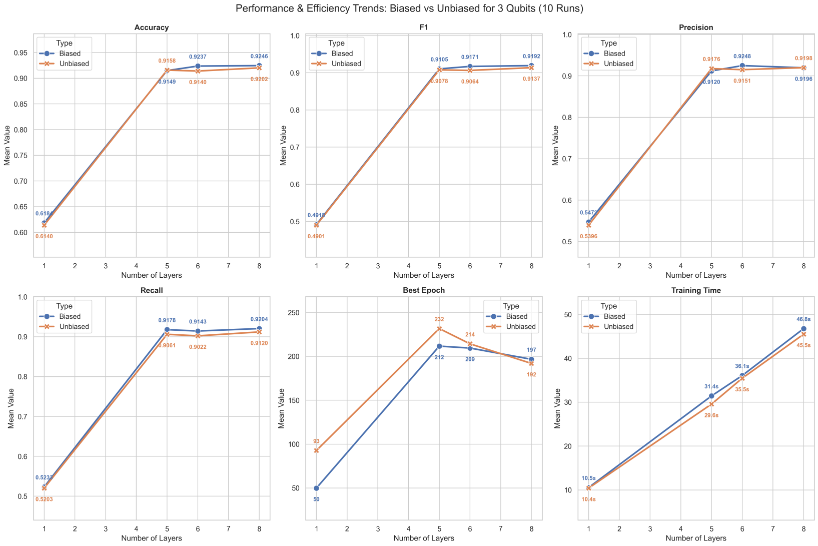


Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 3 Qubits

Three qubit results, circuit of best unbiased VQC

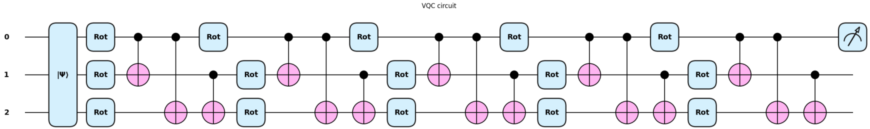


Figure: Circuit of the best 3-qubit unbiased classifier

Three qubit results, loss evolution of best unbiased VQC

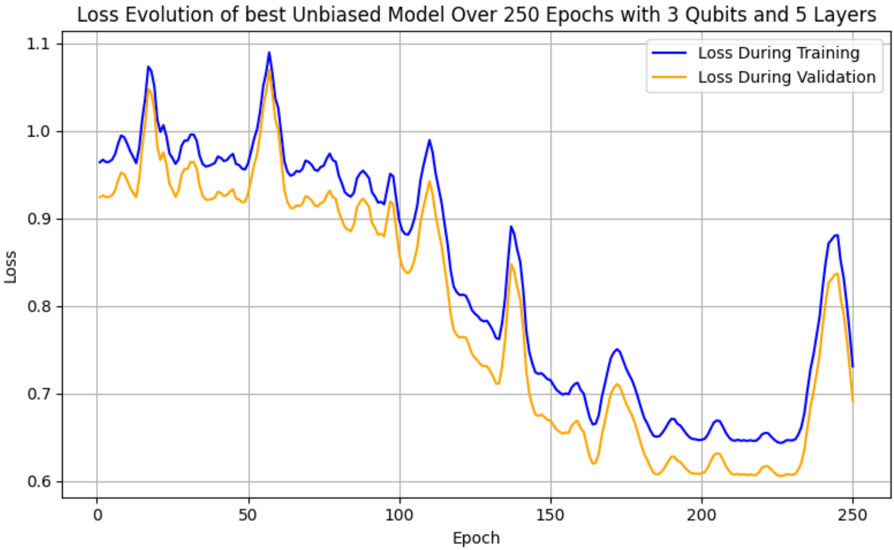


Figure: Evolution of the loss function of the best 3-qubit unbiased classifier

Three qubit results, performance evolution of best unbiased VQC

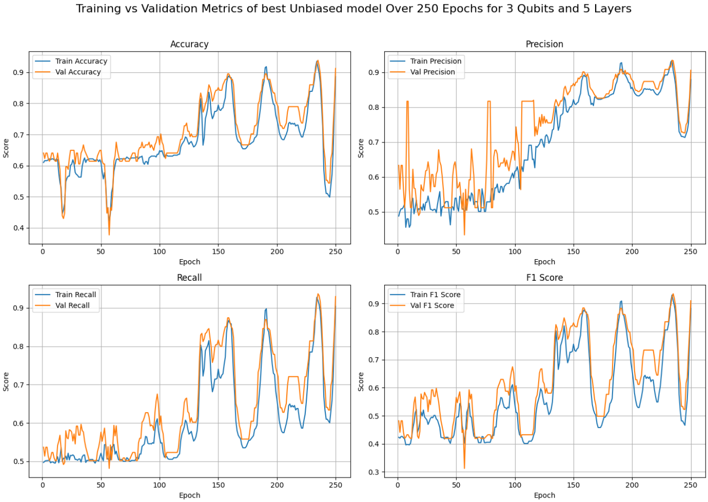


Figure: Validation and Training curves for the 4 key metrics of the best 3-qubit unbiased classifier

Three qubit results, circuit of best biased VQC

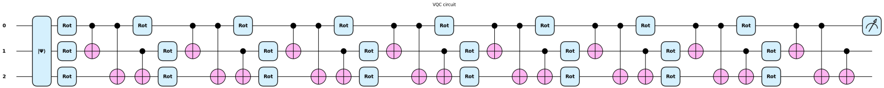
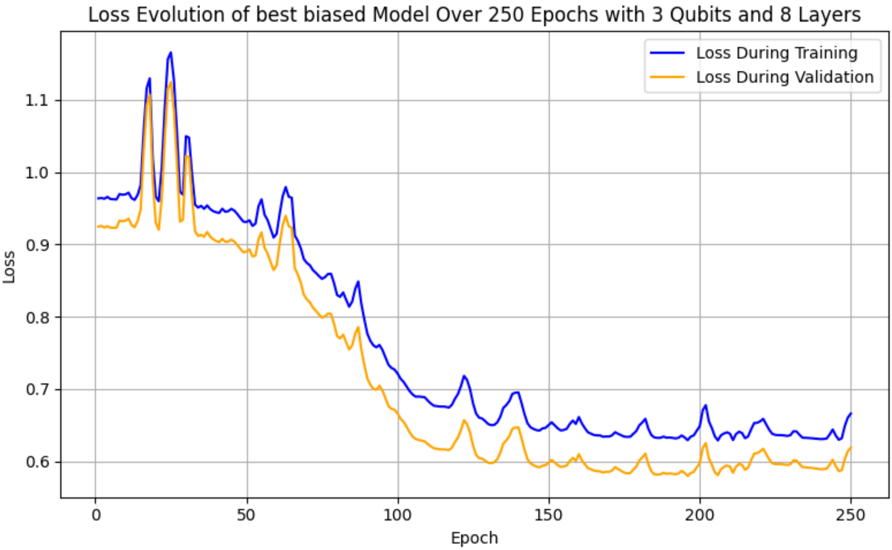


Figure: Circuit of the best 3-qubit biased classifier

Three qubit results, loss evolution of best biased VQC



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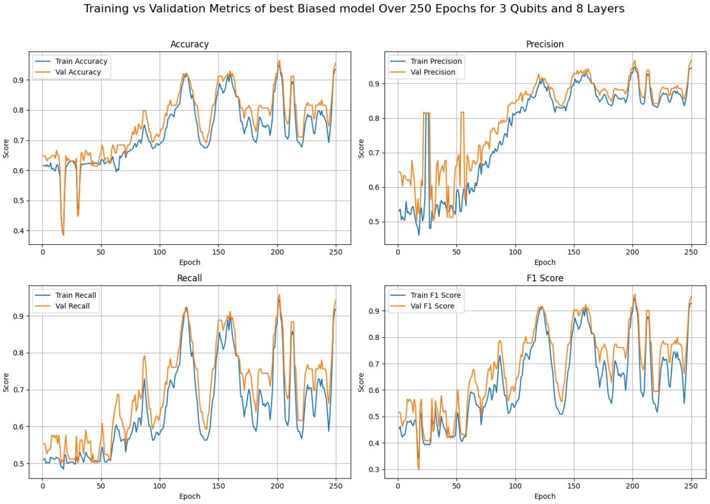

Figure: Validation and Training curves for the 4 key metrics of the best 3-qubit biased classifier

Four qubit results

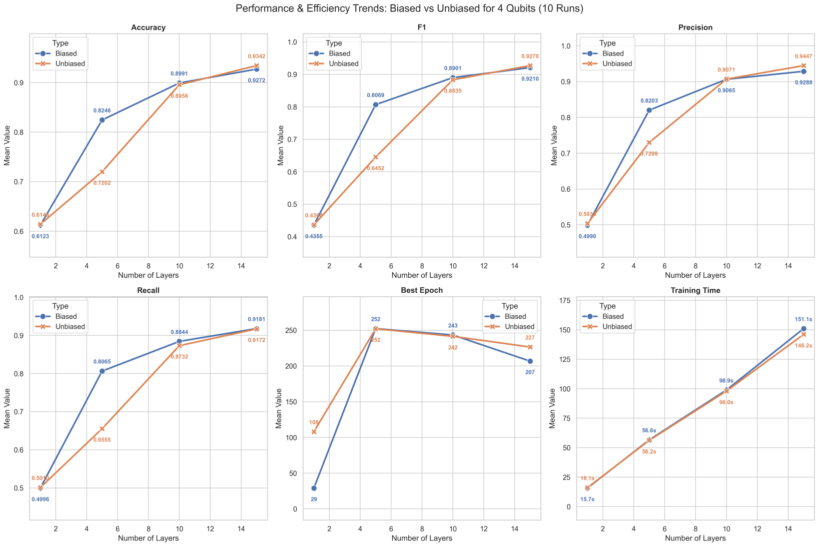

Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 4 Qubits

Four qubit results, circuit of best unbiased VQC

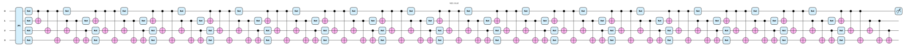

Figure: Circuit of the best 4-qubit unbiased classifier

Four qubit results, loss evolution of best unbiased VQC

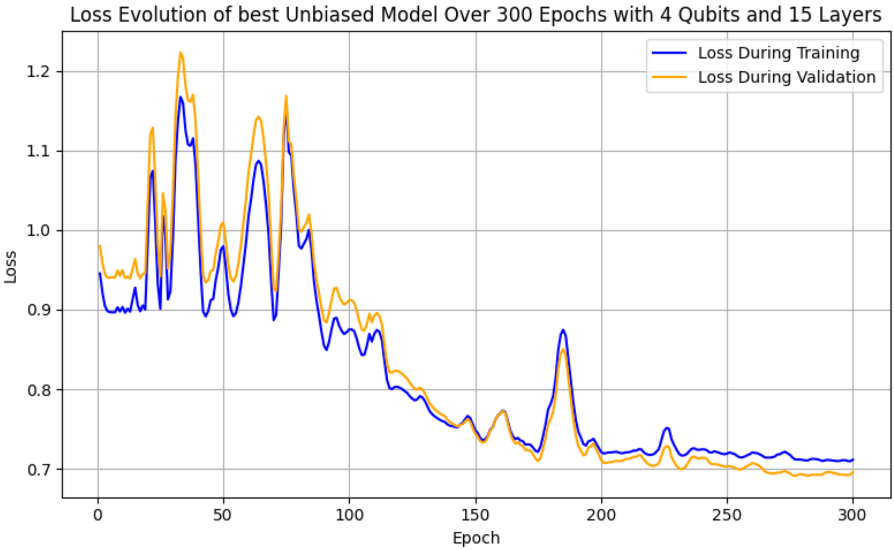

Figure: Evolution of the loss function of the best 4-qubit unbiased classifier

Four qubit results, performance evolution of best unbiased VQC

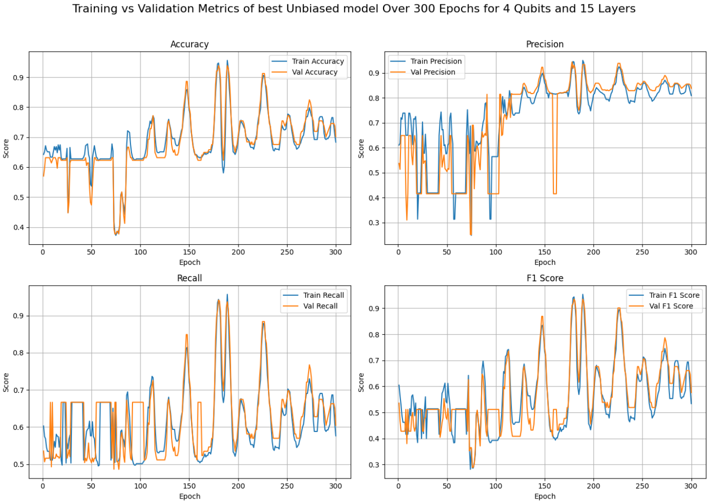

Figure: Validation and Training curves for the 4 key metrics of the best 4-qubit unbiased classifier

Four qubit results, circuit of best biased VQC

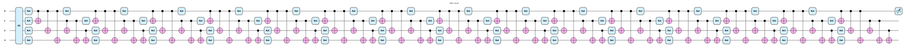

Figure: Circuit of the best 4-qubit biased classifier

Four qubit results, loss evolution of best biased VQC

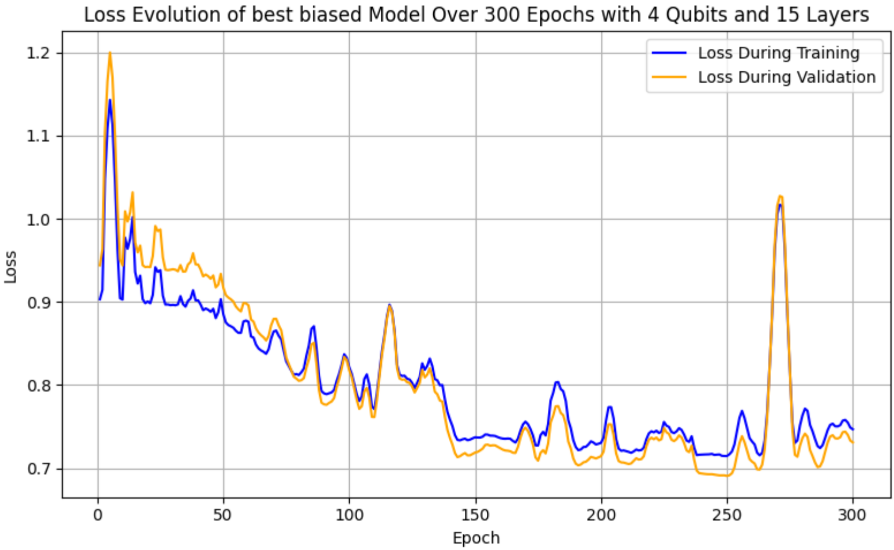

Figure: Evolution of the loss function of the best 4-qubit biased classifier

Four qubit results, performance evolution of best biased VQC

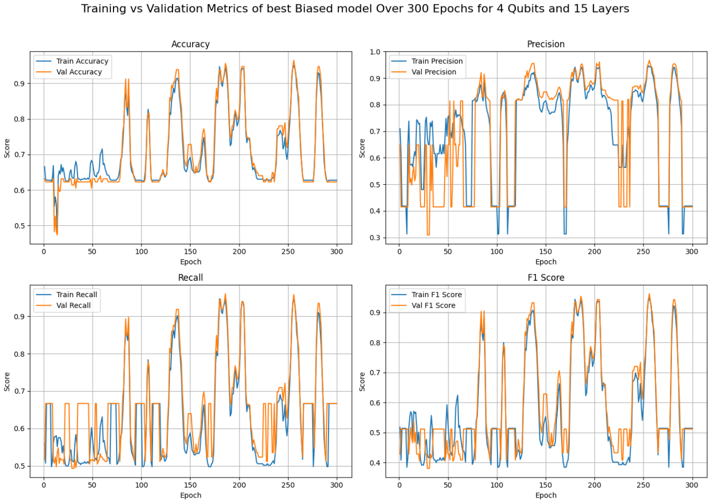

Figure: Validation and Training curves for the 4 key metrics of the best 4-qubit biased classifier

Five qubit results

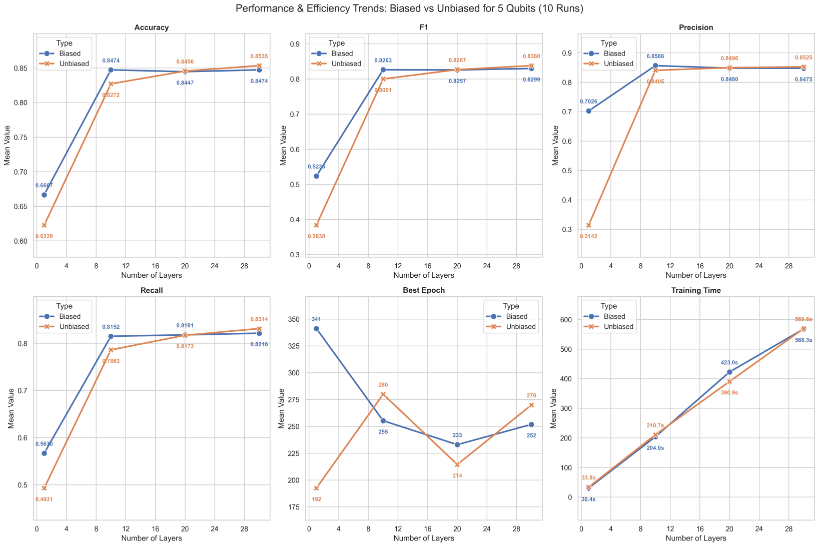

Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 5 Qubits

Five qubit results, circuit of best unbiased VQC

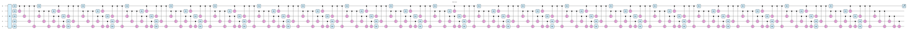

Figure: Circuit of the best 5-qubit unbiased classifier

Five qubit results, loss evolution of best unbiased VQC

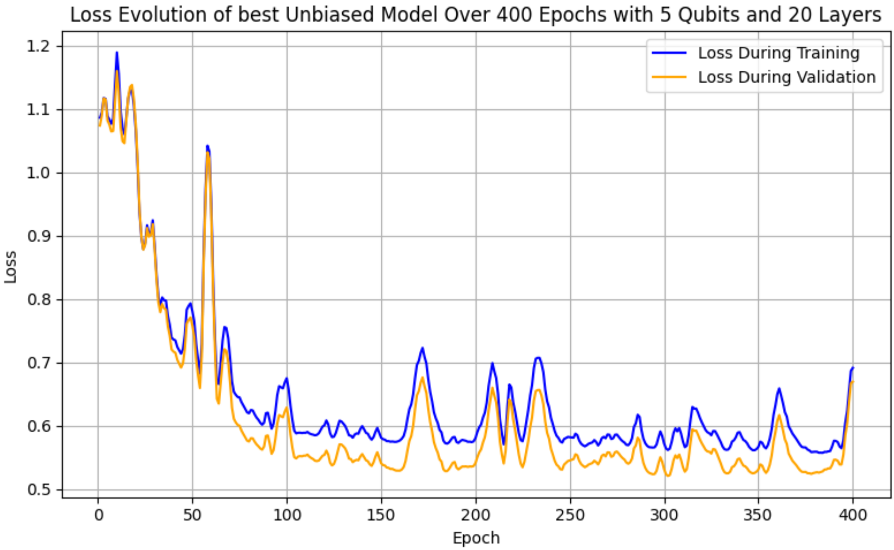

Figure: Evolution of the loss function of the best 5-qubit unbiased classifier

Five qubit results, performance evolution of best unbiased VQC

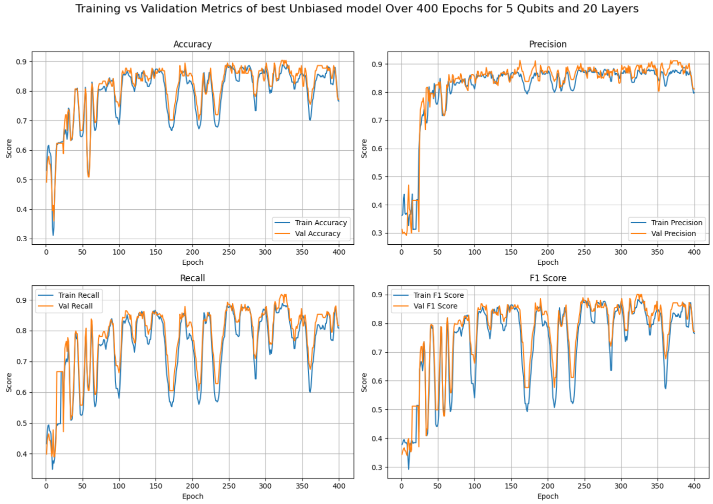

Figure: Validation and Training curves for the 4 key metrics of the best 5-qubit unbiased classifier

Five qubit results, circuit of best biased VQC

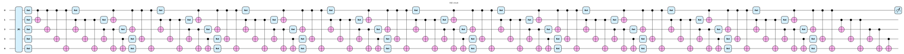

Figure: Circuit of the best 5-qubit biased classifier

Five qubit results, loss evolution of best biased VQC

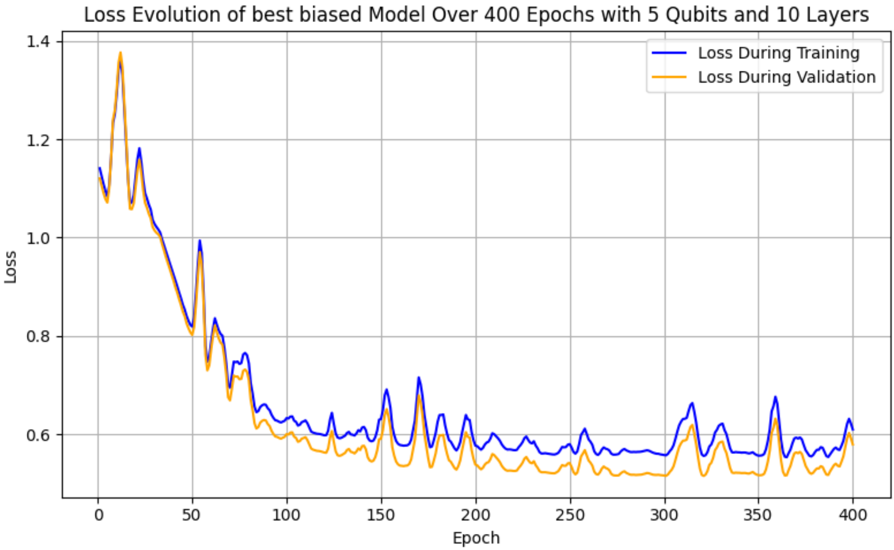

Figure: Evolution of the loss function of the best 5-qubit biased classifier

Five qubit results, performance evolution of best biased VQC

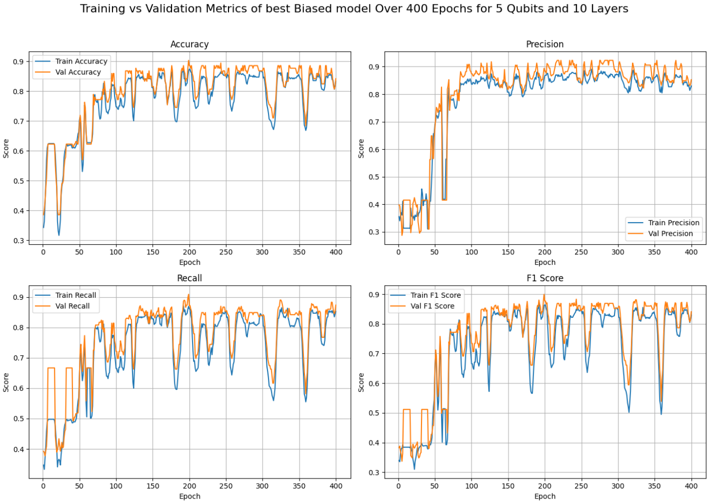

Figure: Validation and Training curves for the 4 key metrics of the best 5-qubit biased classifier

Comparison, Best unbiased performers

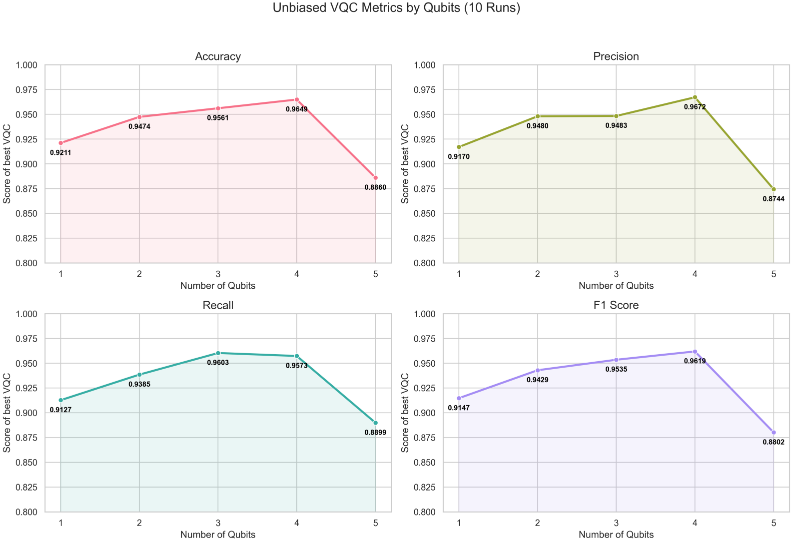

Figure: Performance of best unbiased classifier per number of qubits across all runs

Comparison, Best biased performers

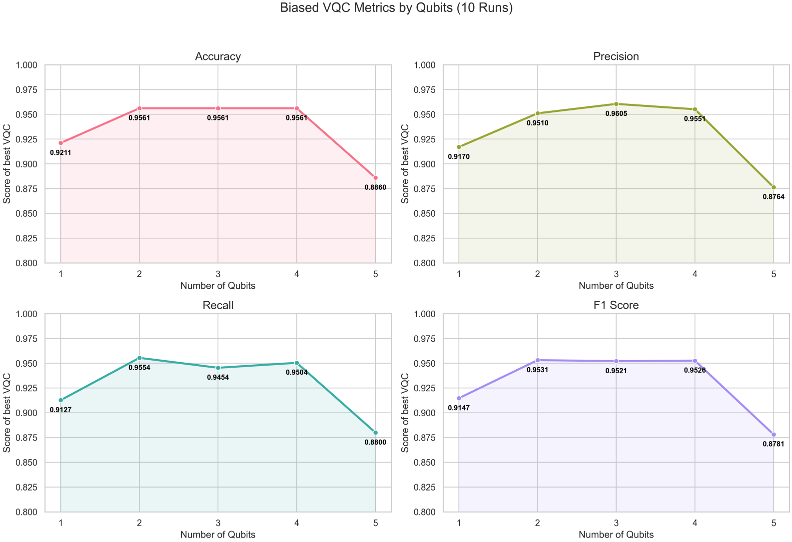

Figure: Performance of best biased classifier per number of qubits across all runs

Comparison, parameters needed for best performers

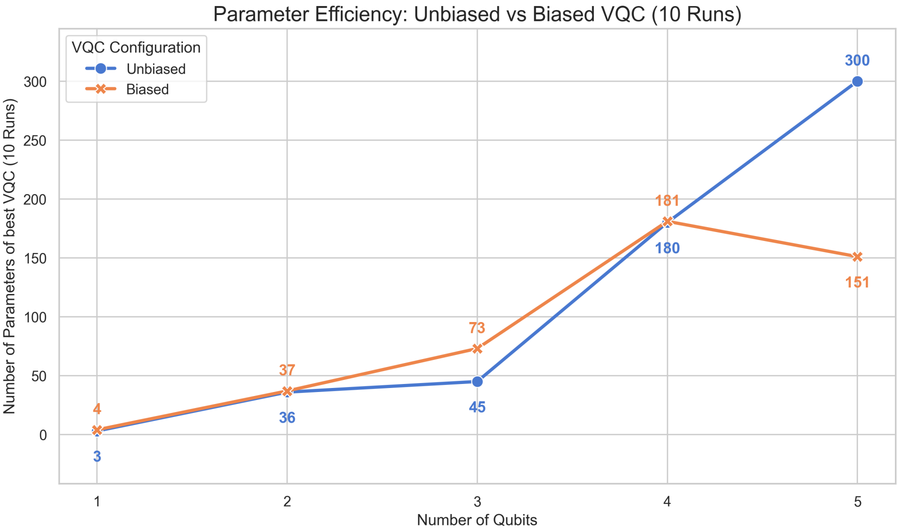

Figure: Number of trainable parameters in respect to the number of qubits for biased and unbiased classifiers

VQCs vs Logistic Regression

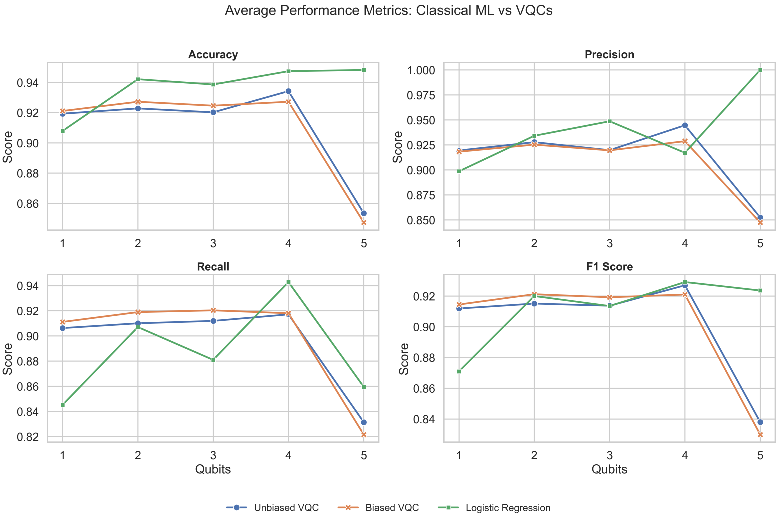

Figure: Average performance of the Quantum Models and the classical Logistic Regression model across 10 different runs for all the different number of qubits

Conclusion

VQCs show **strong potential**, especially in low dimensionality problems

Classical ML still **slightly** superior

Future quantum hardware will **improve** results

Thank you for your attention!

Questions?